

Labwork 2 Report: Get to Know Your GPU

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1 Introduction

In this labwork, I used the `numba.cuda` Python library to query and display the hardware information of my local GPU.

2 Implementation

I wrote a script to detect the GPU, query its specs, and compute the total CUDA cores based on the multiprocessor count.

Below is the pseudocode of the implementation:

Algorithm: Query GPU Specs

If CUDA **is not** available:

 Print "CUDA is not available" **and** exit

Detect **and** select GPU device (ID = 0)

Get Device Name **and** Device ID

Get Multiprocessor Count (SM)

Total Cores = SM Count * 128 (GTX 1050)

Get Total Memory **and** Free Memory **in** Bytes

Convert Memory to GB (bytes / 1024^3)

Print Device ID, Name, SM Count, Total Cores, **and** Memory

3 Results

Executing the program on my machine printed the following GPU specs:

- **Device ID:** 0
- **Device Name:** NVIDIA GeForce GTX 1050
- **Multiprocessors (SM):** 6
- **Cores per SM:** 128
- **Total CUDA Cores:** $6 \times 128 = 768$
- **Total Memory:** 2.94 GB (3,158,114,304 bytes)
- **Free Memory:** 2.89 GB (3,100,508,160 bytes)

4 Conclusion

The script executed successfully. My local GPU is an NVIDIA GeForce GTX 1050 with 6 Streaming Multiprocessors, 768 CUDA cores, and approximately 2.94 GB of VRAM.